What is npm?

npm Overview:

- npm stands for Node Package Manager and is the world's largest software library (registry).
- It is also a software package manager and installer.

The World's Largest Software Registry:

- npm is the world's largest software registry, containing over 800,000 code packages.
- Open-source developers use npm to share software, and many organizations also use npm to manage private development.

Using npm is Free:

- npm is free to use.
- You can download all public software packages from npm without any registration or login required.

Command Line Client:

- npm includes a CLI (Command Line Interface) that can be used to download and install software.

  Windows Example:

  ```
  C:\> npm install <package>
  ```

  Mac OS Example:

  ```
  > npm install <package>
  ```

Installing npm:

- npm is installed with Node.js.
- This means you need to install Node.js to get npm installed on your computer.
- Download Node.js from the official website: https://nodejs.org

Software Package Manager:

- The name npm originally stood for Node Package Manager when it was first created as a package manager for Node.js.
- All npm packages are defined in files called `package.json`.
- The content of `package.json` must be written in JSON.
- At least two fields must be present in the definition file: `name` and `version`.

Example of package.json:

```
{
  "name": "foo",
  "version": "1.2.3",
  "description": "A package for fooing things",
  "main": "foo.js",
  "keywords": ["foo", "fool", "foolish"],
  "author": "John Doe",
  "license": "ISC"
}
```

Managing Dependencies:

- npm can manage dependencies.
- It can install all the dependencies of a project in one command.
- Dependencies are defined in `package.json`.

Sharing Your Software:

- If you want to share your own software in the npm registry, sign in at: https://www.npmjs.com

Publishing a Package:

- You can publish any directory from your computer as long as the directory contains a `package.json` file.

  Check if npm is installed:

  ```
  C:\> npm
  ```

  Check if you are logged in:

  ```
  C:\> npm whoami
  ```

  If not, log in:

  ```
  C:\> npm login
  Username: <your username>
  Password: <your password>
  ```

  Navigate to your project and publish your project:

  ```
  C:\Users\myuser>cd myproject
  C:\Users\myuser\myproject>npm publish
  ```